NEW ERA PUBLIC SCHOOL
Midterm Examination (2026-27)
Class: X
Subject: Science

M.M: 80

Time: 3 Hours.

General Instructions:
(i) This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.
(ii) All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.

SECTION A

1. Which of the following is NOT a component of the human digestive system? (1)
(a) Liver (b) Pancreas (c) Trachea (d) Duodenum
2. The pigment present in the mesophyll cells that traps solar energy for photosynthesis is: (1)
(a) Xanthophyll (b) Chlorophyll (c) Carotene (d) Anthocyanin
3. The breakdown of pyruvate to give carbon dioxide, water and energy takes place in (1)
(a) cytoplasm. (c) chloroplast. (b) mitochondria. (d) nucleus.
4. Which of the following is a double circulation feature seen in humans? (1)
(a) Blood passes through the heart once in one cycle
(b) Blood passes through the heart twice in one cycle
(c) Blood does not pass through the heart
(d) Blood passes through the lungs only

In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is NOT the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

5. **Assertion (A)** : Fishes have gills for respiration while terrestrial animals have lungs. (1)
Reason (R) : The amount of dissolved oxygen is fairly low compared to the amount of oxygen in the air.

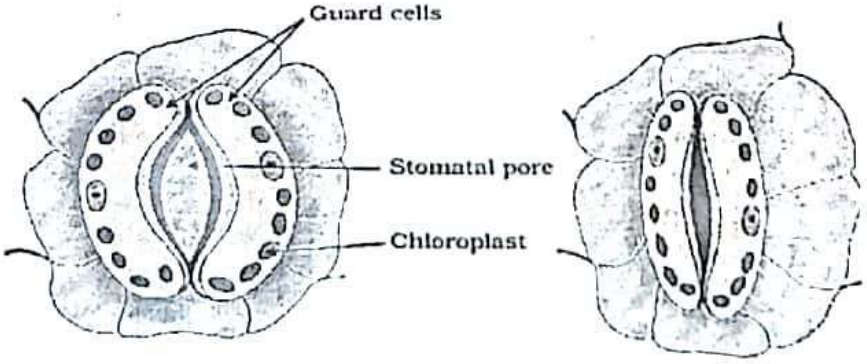
6. **Attempt either A or B** (2)

A. How are water and minerals transported in plants?

OR

B. How is food transported in plants?

7. Answer the following questions based on the diagram given below:



Scanned with CamScanner

(a) State two functions of stomatal pores in plant cells?
(b) Explain the reasons for opening and closing of stomata?
(c) Illustrate what would happen to stomatal pores during hot and dry conditions.

8. Riya was studying the human excretory system and learned that the kidneys filter blood through millions of nephrons. Each nephron has a cup-shaped structure called Bowman's capsule surrounding a cluster of blood capillaries. The filtrate initially formed contains glucose, amino acids, salts, and urea. As the filtrate passes through the tubule, useful substances like glucose and most water are selectively reabsorbed back into the blood, while urea and excess salts pass on to form urine. (4)

(a) Why is selective reabsorption necessary in the nephron? Give two reasons.
(b) If the filtration membrane of the Bowman's capsule gets damaged, which substance might abnormally appear in the urine?
(c) Explain why the amount of urine produced depends on how much excess water is present in the body.

OR

(c) Patients with kidney failure often require dialysis. Explain the basic principle on which a dialysis machine works. (5)

9. **Attempt either A or B**

A. (a) Draw a labelled diagram of the human excretory system showing the kidneys, ureters, urinary bladder and urethra.
(b) Explain the role of the alveoli in gaseous exchange in the human lungs.
(c) Why is haemoglobin essential for the transport of oxygen in human blood?

OR

B. (a) Draw a labelled diagram of the human heart showing all four chambers and the major blood vessels.
(b) Why is a double circulatory system necessary in humans?
(c) Name the blood vessel that carries deoxygenated blood away from the heart.

SECTION B

10. Which type of chemical reaction is represented by: $2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$ (1)
on heating?
(a) Combination reaction
(b) Decomposition reaction
(c) Displacement reaction
(d) Double displacement reaction

11. A reaction in which a more reactive element displaces a less reactive element from its compound is called: (1)
(a) Combination reaction
(b) Decomposition reaction
(c) Displacement reaction
(d) Neutralization reaction

12. A gas is evolved when a metal reacts with a dilute acid? Choose the correct pair of gas and the test to identify it: (1)
(a) Hydrogen, turns lime water milky
(b) Carbon dioxide, extinguishes a burning candle
(c) Hydrogen, burns with a pop sound when a burning candle is brought near it
(d) Nitrogen, Turns moist blue litmus red

13. Which of the following is the general formula for saturated hydrocarbons (alkanes)? (1)
(a) C_nH_{2n} (b) C_nH_{2n-2} (c) C_nH_{2n-2} (d) C_nH_n

Scanned with CamScanner

Vinegar used in the kitchen is a dilute solution of. (1)

(a) Formic acid (b) Citric acid (c) Acetic acid (d) Tartaric acid

Assertion (A) : Milk of magnesia is used as an antacid to treat acidity (1)

Reason (R) : Milk of magnesia is basic in nature and neutralizes the excess acid in the stomach.

a) Both A and R are true and R is the correct explanation of A

b) Both A and R are true but R is NOT the correct explanation of A

c) A is true but R is false

d) A is false but R is true

Balance the following chemical equation and identify the type of reaction: (2)

i) $Fe + H_2O \rightarrow Fe_3O_4 + H_2$

ii) $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + NaCl$

Why does carbon form a very large number of compounds compared to most other elements? Write the reasons for this. (2)

ttempt either A or B (3)

student added a few drops of universal indicator to three test tubes containing dilute HCl, stilled water and dilute NaOH respectively.

) Predict the pH range and the colour change in each test tube.

) What would happen to the pH if the dilute HCl is gradually diluted further with water?

OR

) Write the chemical formula and chemical name of Plaster of Paris

) How is Plaster of Paris manufactured?

Why should Plaster of Paris be stored in moister proof containers?

shiny brown colored element 'X' on heating in air becomes black in color. (3)

hen hydrogen gas is passed over this heated black substance, it regains its original shiny wn color.

Identify element 'X' and the black colored substance formed.

Write balanced chemical equations for both reactions involved.

's mother was preparing food. She added a substance to the vegetable curry that ed out too sour due to excess use of tamarind, and observed bubbling when a pinch king soda was added.

Vhat type of reaction takes place when baking soda is added to an acidic substance? (1)

entify the gas produced and write a test to confirm it. (1)

hat happens to the baking soda when it is heated along with the food? (2)

OR

w is baking soda prepared? Write a balanced chemical equation. (2)

at happens when ethanol reacts with sodium metal? Write the chemical equation. (5)

y does the addition of ethanol to petrol help reduce vehicular pollution?

ne the functional group present in ethanoic acid.

at is the industrial name for ethanoic acid?

OR

w the electron dot structure of (i) ethane (ii) ammonia

t is meant by a homologous series? Write the next two homologous for Propanone

e the name of the following carbon compounds

H3CH2CH2CH2-Br (ii) HCOOH

Scanned with CamScanner

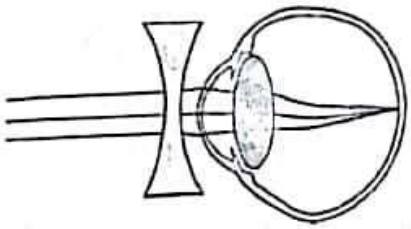
SECTION C

22. The image formed by a concave mirror is found to be virtual, erect and larger than the object. The position of the object must be: (1)
(a) At the centre of curvature
(b) Beyond the centre of curvature
(c) Between the pole and the focus
(d) At infinity

23. A ray of light travelling from a rarer medium to a denser medium: (1)
(a) Bends away from the normal
(b) Bends towards the normal
(c) Passes undeviated
(d) Gets completely reflected

24. While observing the human eye, a student notes that when stepping out from a dimly lit room into bright noon sunlight, the eye adjusts itself to prevent glare. Which specific sequence of muscular and anatomical actions occurs during this transition? (1
(a) Ciliary muscles relax ----> Eye lens becomes thin ----> Focal length increases.
(b) Iris contracts ---> Pupil size decreases ----> Amount of light entering the retina reduces.
(c) Iris expands ----> Pupil size increases ----> Photoreceptor rods are deactivated.
(d) Ciliary muscles contract ----> Eye lens becomes thick ----> Focal length decreases.

25. Identify the correct defect of vision and the corrective lens shown in the figure below



(a) Hypermetropia, concave lens
(b) Myopia, convex lens
(c) Myopia, bifocal lens
(d) Myopia, concave lens

26. Three resistors of resistances 2 Ω, 3 Ω, and 6 Ω are connected in parallel. What is the total equivalent resistance of the combination?
(a) 11 Ω (b) 1 Ω (c) 5 Ω (d) 2 Ω

27. Which of the following correctly represents the direction of the magnetic field lines around a straight current-carrying conductor?
(a) Straight lines parallel to the conductor
(b) Concentric circles in a plane perpendicular to the conductor
(c) Radial lines from the conductor
(d) Straight lines perpendicular to the conductor

28. A magnetic compass needle placed near a current-carrying wire gets deflected. This effect was first observed by: (1
(a) Faraday (b) Newton (c) Oersted (d) Ampere

29. **Assertion (A)** : The refractive index of diamond is much higher than that of glass. (1
Reason (R) : The speed of light in diamond is greater than the speed of light in glass.
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is NOT the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

30. **Assertion (A)** : A soft iron core is used in an electromagnet. (1)
Reason (R) : Soft iron gets magnetised only in the presence of an external magnetic field and loses its magnetism once the field is removed.
(a) Both A and R are true and R is the correct explanation of A

Scanned with CamScanner

(b) Both A and R are true but R is NOT the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

31. **Attempt either A or B**

A. An electric iron of resistance 20 Ω takes a current of 5 A. Calculate the heat developed in 30 s. (2)

OR

B. An electric heater draws a current of 5 A from a 220 V supply. Calculate the power of the heater and the electrical energy consumed in 2 hours.

32. Draw the direction of the magnetic field produced around a straight current-carrying conductor. How does the strength of this magnetic field depend on the current and the distance from the conductor? (2)

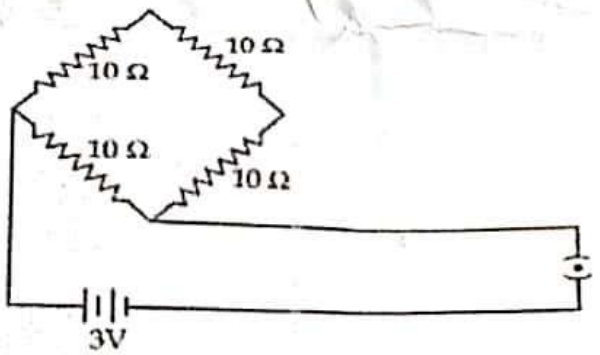
33. A 2 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 10 cm at a distance of 15 cm from the lens. Calculate the image distance and the size of the image. (2

34. The near point of vision recedes with age in some people. (3
(a) Name the defect and the type of lens used to correct it.
(b) What is the value of near point and far point of a normal adult human being?

35. Two lenses of power +5 D and −2.5 D are placed in contact. Find the focal length and power of the combination, and state whether it behaves as a converging or diverging system. (2

36. Derive an expression for a combination of resistors connected in series combination. (

37. Find the current drawn from the battery by the network of four resistors shown in the figure. (



38. Priya performed an experiment in the school lab. She wound an insulated copper wire aro a cylindrical cardboard tube to form a solenoid and connected its ends to a battery, key, a rheostat in series. On passing current, she brought a compass needle near one end and observed a deflection. When she reversed the direction of current using a commutator, tl deflection of the compass reversed too. She then inserted a soft iron rod inside the solen and noticed that the compass needle deflected much more strongly than before.
(a) What is a solenoid?
(b) State the role of the soft iron core for the solenoid.
(c) List two ways by which the magnetic field produced by a solenoid can be made stronger.

OR

(c) Draw a diagram to show the magnetic field lines around a current carrying solenoic

Scanned with CamScanner

39. (a) Why the colour of the sky is blue? Will it be same to an astronaut? Explain. (5)
(b) Define the power of accommodation of the eye.
(c) Write the functions of the following (i) pupil (ii) optic nerve

OR

(a) Draw a ray diagram to show the refraction of a monochromatic ray of light through (5)
A glass prism and label the following
(i) angle of deviation (ii) refracted ray (iii) emergent ray (iv) angle of emergence
(b) What is presbyopia? Why does it occur, and how is it different from hypermetropia?

Scanned with CamScanner